Cognizant - UpGrad Training Program
WEEKLY CODING SUBMISSION
Name : Durgesh Vijay Bhangale
Week : Week 1
Trainer : Kunal

Problem 1
Assessment Goal: Check if learners understand basic HTML structure and content creation.

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>My First HTML Page</title>
  <meta charset="UTF-8">
</head>
<body>

  <h1>Welcome to My Page</h1>
  <p>Hello, my name is Durgesh. I am learning HTML fundamentals and this is my basic
webpage.</p>

  <h2>My Hobbies</h2>
  <ul>
    <li>Reading</li>
    <li>Coding</li>
    <li>Playing Cricket</li>
    <li>Listening to Music</li>
  </ul>

  <h2>My Daily Routine</h2>
  <ol>
    <li>Wake up early</li>
    <li>Exercise</li>
    <li>Study and practice coding</li>
    <li>Work on projects</li>
    <li>Relax and sleep</li>
  </ol>

  <h2>Student Marks Table</h2>
  <table border="1" cellpadding="5" cellspacing="0">
    <tr>
      <th>Student Name</th>
      <th>Subject</th>
      <th>Marks</th>
    </tr>
```

```
      <tr>
        <td>Durgesh</td>
        <td>HTML Fundamentals</td>
        <td>95</td>
      </tr>
    </table>
</body>
</html>
```

Problem 2: Restaurant Menu Webpage (Level-1)

CODE :

```
<!DOCTYPE html>
<html>
<head>
   <title>My Restaurant Menu</title>
   <meta charset="UTF-8">
</head>
<body>

   <h1>Delicious Bites</h1>

   <p>Welcome to Delicious Bites! We serve a variety of tasty dishes prepared with love and care.
Explore our menu below.</p>

   <h2>Menu Categories</h2>
   <ul>
      <li>Main Course</li>
      <li>Breakfast</li>
      <li>Beverages</li>
      <li>Desserts</li>
   </ul>

   <h2>Price List</h2>
```

```
  <table border="1" cellpadding="5" cellspacing="0" title="Restaurant Menu" align="center">
    <tr>
      <th>Item Name</th>
      <th>Category</th>
      <th>Price (₹)</th>
    </tr>
    <tr>
      <td>Paneer Butter Masala</td>
      <td>Main Course</td>
      <td>220</td>
    </tr>
    <tr>
      <td>Veg Biryani</td>
      <td>Main Course</td>
      <td>180</td>
    </tr>
    <tr>
      <td>Masala Dosa</td>
      <td>Breakfast</td>
      <td>90</td>
    </tr>
    <tr>
      <td>Cold Coffee</td>
      <td>Beverages</td>
      <td>120</td>
    </tr>
  </table>
</body>
</html>
```

Problem 3: Personal Grocery Checklist (Level-1)

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>Weekly Grocery Checklist</title>
  <meta charset="UTF-8">
</head>
<body>
```

```
  <h1>Weekly Grocery Checklist</h1>

  <p>Hello, I am Durgesh. This is my simple webpage to plan weekly grocery shopping.</p>

  <h2>High Priority Items</h2>
  <ol title="Essential Groceries">
    <li>Rice</li>
    <li>Milk</li>
    <li>Vegetables</li>
    <li>Cooking Oil</li>
  </ol>

  <h2>Optional Items</h2>
  <ul title="Non-Essential Groceries">
    <li>Snacks</li>
    <li>Ice Cream</li>
    <li>Soft Drinks</li>
  </ul>
</body>
</html>
```

Practice Question 4: Basic Personal Webpage (HTML Structure + Elements)

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>About Me Page</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

  <h1>Durgesh Bhangale</h1>

  <p>
    Hello! My name is Durgesh Bhangale. I am passionate about learning
    <mark>technology</mark> and building creative solutions.
    I also enjoy <b>coding</b> and exploring new ideas.
  </p>

  <h2>My Goals</h2>
  <ul>
    <li>Improve my programming skills</li>
    <li>Contribute to open-source projects</li>
    <li>Build impactful data-driven applications</li>
    <li>Enhance problem-solving abilities</li>
```

```
    <li>Secure a rewarding career in technology</li>
  </ul>

  <footer>
    <p>Created by Durgesh Bhangale - 2026</p>
  </footer>
</body>
</html>
```

Practice Question 5: Student Marks Table + Ordered List (Table + Rows + Headers)

CODE :

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Marks Report</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

  <h1>Student Marks Report</h1>
```

```
<table border="1" cellpadding="8" cellspacing="0" align="center" title="Marks Report">
  <tr>
    <th>Roll No</th>
    <th>Name</th>
    <th>Subject</th>
    <th>Total Marks</th>
    <th>Marks Obtained</th>
    <th>Result</th>
  </tr>
  <tr>
    <td>01</td>
    <td>Durgesh</td>
    <td>HTML Fundamentals</td>
    <td>100</td>
    <td>95</td>
    <td style="color:green;">Pass</td>
  </tr>
  <tr>
    <td>02</td>
    <td>Ravi</td>
    <td>CSS Basics</td>
    <td>100</td>
    <td>60</td>
    <td style="color:green;">Pass</td>
  </tr>
  <tr>
    <td>03</td>
    <td>Priya</td>
    <td>JavaScript</td>
    <td>100</td>
    <td>40</td>
    <td style="color:red;">Fail</td>
  </tr>
  <tr>
    <td>04</td>
    <td>Neha</td>
    <td>Python</td>
    <td>100</td>
    <td>85</td>
    <td style="color:green;">Pass</td>
  </tr>
  <tr>
    <td>05</td>
    <td>Amit</td>
    <td>Data Science</td>
    <td>100</td>
    <td>30</td>
    <td style="color:red;">Fail</td>
```

```
      </tr>
    </table>

    <hr>

    <h2>Top 5 Subjects in My Course</h2>
    <ol>
      <li>HTML Fundamentals</li>
      <li>CSS Basics</li>
      <li>JavaScript</li>
      <li>Python</li>
      <li>Data Science</li>
    </ol>
</body>
</html>
```

Practice Question 6: Restaurant Menu Page (Semantic Tags + Table + Lists + Links)

CODE :

```
<!DOCTYPE html>
<html>
<head>
    <title>Restaurant Menu</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

    <header>
      <h1>Durgesh's Delight</h1>
      <p>Serving happiness on every plate!</p>
    </header>
```

```
<nav>
  <a href="#" title="Go to Home">Home</a> |
  <a href="#" title="View Menu">Menu</a> |
  <a href="#" title="Contact Us">Contact</a>
</nav>

<section>
  <h2>Menu Items</h2>
  <table border="1" cellpadding="8" cellspacing="0" align="center" title="Restaurant Menu">
    <tr>
      <th>Item Name</th>
      <th>Category</th>
      <th>Price (₹)</th>
      <th>Availability</th>
    </tr>
    <tr>
      <td>Paneer Tikka</td>
      <td>Starter</td>
      <td>150</td>
      <td>Available</td>
    </tr>
    <tr>
      <td>Veg Biryani</td>
      <td>Main Course</td>
      <td>200</td>
      <td>Available</td>
    </tr>
    <tr>
      <td>Butter Naan</td>
      <td>Main Course</td>
      <td>40</td>
      <td>Available</td>
    </tr>
    <tr>
      <td>Chocolate Cake</td>
      <td>Dessert</td>
      <td>120</td>
      <td>Available</td>
    </tr>
    <tr>
      <td>Cold Coffee</td>
      <td>Drinks</td>
      <td>90</td>
      <td>Available</td>
    </tr>
    <tr>
      <td>Fruit Salad</td>
      <td>Dessert</td>
```

```
            <td>80</td>
            <td>Available</td>
          </tr>
        </table>
        <h3>Top 5 Recommended Items</h3>
        <ol>
          <li>Paneer Tikka</li>
          <li>Veg Biryani</li>
          <li>Chocolate Cake</li>
          <li>Cold Coffee</li>
          <li>Butter Naan</li>
        </ol>

        <h3>Available Offers</h3>
        <ul>
          <li>Buy 1 Get 1 Free on Cold Coffee</li>
          <li>20% Off on Veg Biryani</li>
          <li>Free Dessert with Main Course</li>
          <li>Weekend Special Combo at ₹299</li>
          <li>Student Discount: 10% Off</li>
        </ul>
    </section>
    <footer>
        <p>Thank you for visiting!</p>
        <p>Contact us: <a
href="mailto:durgeshbhangale12@gmail.com">durgeshbhangale12@gmail.com</a></p>
    </footer>
</body>
</html>
```

DAY 2

Problem 1

Assessment Goal: Validate understanding of links and form elements.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Home Page</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
  <!-- Navigation Bar -->
  <nav>
    <a href="index.html">Home</a> |
    <a href="contact.html">Contact</a>
  </nav>

  <!-- Content -->
  <h1>Welcome to Durgesh's Website</h1>
  <p>This is the home page. Use the navigation bar to go to the Contact page and register.</p>
</body>
</html>
```

Contact.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Contact Page</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

  <nav>
    <a href="index.html">Home</a> |
    <a href="contact.html">Contact</a>
  </nav>

  <h1>Registration Form</h1>
  <form>

    <label for="name">Name:</label>
```

```
    <input type="text" id="name" name="name" placeholder="Enter your name" required>
    <br><br>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email" placeholder="Enter your email" required>
    <br><br>

    <label for="password">Password:</label>
    <input type="password" id="password" name="password" placeholder="Enter your password"
required>
    <br><br>

    <label>Gender:</label>
    <input type="radio" id="male" name="gender" value="Male" required> Male
    <input type="radio" id="female" name="gender" value="Female"> Female
    <br><br>

    <label for="course">Select Course:</label>
    <select id="course" name="course" required>
      <option value="">--Choose a course--</option>
      <option value="HTML">HTML Fundamentals</option>
      <option value="CSS">CSS Basics</option>
      <option value="JavaScript">JavaScript</option>
      <option value="Python">Python</option>
      <option value="DataScience">Data Science</option>
    </select>
    <br><br>

    <input type="submit" value="Register">
  </form>
</body>
</html>
```

Problem 2: Simple Website Navigation System (Level-1)

1.Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Home - Durgesh Institute</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

  <nav>
```

```
      <a href="index.html">Home</a> |
      <a href="courses.html">Courses</a> |
      <a href="contact.html">Contact</a>
   </nav>

   <h1>Welcome to Durgesh Institute</h1>
   <p>We provide quality education and training in technology.</p>
   <p><a href="#about">Learn more about us</a></p
   <section id="about">
      <h2>About Us</h2>
      <p>Durgesh Institute is dedicated to helping students achieve their career goals.</p>
   </section>
   <p>Follow us on
      <a href="https://www.facebook.com" target="_blank">Facebook</a>
   </p>
</body>
</html>
```

2.Courses.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Courses - Durgesh Institute</title>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

   <nav>
      <a href="index.html">Home</a> |
      <a href="courses.html">Courses</a> |
      <a href="contact.html">Contact</a>
   </nav>

   <h1>Our Courses</h1>
   <ul>
      <li>HTML Fundamentals</li>
      <li>CSS Basics</li>
      <li>JavaScript Essentials</li>
      <li>Python Programming</li>
      <li>Data Science</li>
   </ul>
</body>
</html>
```

3.Contact.html

```
<!DOCTYPE html>
```

```html
<html>
<head>
   <title>Contact - Durgesh Institute</title>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

   <nav>
      <a href="index.html">Home</a> |
      <a href="courses.html">Courses</a> |
      <a href="contact.html">Contact</a>
   </nav>

   <h1>Contact Us</h1>
   <p>Email: durgesh@gmail.com</p>
   <p>Phone: +91-9999999999</p>
</body>
</html>
```

Problem 3: Student Registration Form with Input Types (Level-1)

Code:

```html
<!DOCTYPE html>
<html>
<head>
   <title>Student Registration Form</title>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
   <h1>Student Registration Form</h1>

   <form>

      <label for="name">Full Name:</label>
      <input type="text" id="name" name="name" placeholder="Enter your name" required>
      <br><br>

      <label for="email">Email Address:</label>
      <input type="email" id="email" name="email" placeholder="Enter your email" required>
      <br><br>

      <label for="phone">Phone Number:</label>
      <input type="tel" id="phone" name="phone"
           placeholder="e.g. 9876543210"
           pattern="[0-9]{10}"
           required>
```

```
      <small>(Enter 10 digits)</small>
      <br><br>

      <label for="age">Age:</label>
      <input type="number" id="age" name="age" min="18" max="60" placeholder="18-60"
required>
      <br><br>


      <label for="dob">Date of Birth:</label>
      <input type="date" id="dob" name="dob" required>
      <br><br>

      <input type="submit" value="Register">
   </form>
</body>
</html>
```

Problem 4: E-Commerce Product Website (Hierarchical Linking + Form + Breadcrumb)
Code :
Index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Home - Durgesh Store</title>
   <meta charset="UTF-8">
   <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

   <nav>
      <a href="index.html">Home</a> |
      <a href="electronics.html" id="electronics-link">Electronics</a> |
      <a href="checkout.html">Checkout</a>
   </nav>

   <h1>Welcome to Durgesh Store</h1>
   <p><a href="#categories">Go to Categories</a></p>
   <p><a href="#offers">Go to Offers</a></p>

   <section id="categories">
      <h2>Categories</h2>
      <ul>
         <li><a href="electronics.html">Electronics</a></li>
         <li>Fashion</li>
         <li>Grocery</li>
         <li>Sports</li>
```

```
    </ul>
  </section>

  <section id="offers">
    <h2>Special Offers</h2>
    <ol>
      <li>10% off on Electronics</li>
      <li>Buy 1 Get 1 Free on Fashion</li>
      <li>Flat ₹100 off Grocery orders above ₹500</li>
      <li>20% off Sports equipment</li>
      <li>Free delivery on orders above ₹999</li>
    </ol>
  </section>

  <p>Follow us on <a href="https://www.facebook.com" target="_blank">Facebook</a></p>
</body>
</html>
```

mobiles.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Mobiles - Durgesh Store</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>

  <nav>
    <a href="index.html">Home</a> |
    <a href="electronics.html" id="electronics-link">Electronics</a> |
    <a href="checkout.html">Checkout</a>
  </nav>

  <p>Home > Electronics > Mobiles</p>

  <h1>Mobile Phones</h1>

  <table border="1" cellpadding="8" cellspacing="0">
    <tr>
      <th>Mobile Name</th>
      <th>Price (₹)</th>
      <th>Stock</th>
      <th>Rating</th>
    </tr>
    <tr>
      <td>Samsung Galaxy</td>
```

```
        <td>15000</td>
        <td><meter value="70" min="0" max="100"></meter></td>
        <td>★★★★☆</td>
      </tr>
      <tr>
        <td>iPhone 13</td>
        <td>60000</td>
        <td><meter value="30" min="0" max="100"></meter></td>
        <td>★★★★★</td>
      </tr>
      <tr>
        <td>Redmi Note</td>
        <td>12000</td>
        <td><meter value="90" min="0" max="100"></meter></td>
        <td>★★★☆☆</td>
      </tr>
      <tr>
        <td>OnePlus Nord</td>
        <td>25000</td>
        <td><meter value="50" min="0" max="100"></meter></td>
        <td>★★★★☆</td>
      </tr>
      <tr>
        <td>Realme X</td>
        <td>18000</td>
        <td><meter value="20" min="0" max="100"></meter></td>
        <td>★★★☆☆</td>
      </tr>
    </table>

    <!-- Progress Example -->
    <h2>Order Processing</h2>
    <progress value="50" max="100"></progress> 50% Processing
</body>
</html>
```

3.electronics.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Electronics - Durgesh Store</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
</head>
<body>
```

```
    <nav>
      <a href="index.html">Home</a> |
      <a href="electronics.html" id="electronics-link">Electronics</a> |
      <a href="checkout.html">Checkout</a>
    </nav>

    <p>Home > Electronics</p>

    <h1>Electronics Section</h1>

    <table border="1" cellpadding="8" cellspacing="0">
      <tr>
        <th>Product Type</th>
        <th>Items Available</th>
        <th>Discount</th>
      </tr>
      <tr>
        <td>Mobiles</td>
        <td>50</td>
        <td>10%</td>
      </tr>
      <tr>
        <td>Laptops</td>
        <td>20</td>
        <td>15%</td>
      </tr>
      <tr>
        <td>Televisions</td>
        <td>30</td>
        <td>20%</td>
      </tr>
      <tr>
        <td>Headphones</td>
        <td>100</td>
        <td>5%</td>
      </tr>
    </table>

    <h2>Explore Categories</h2>
    <ul>
      <li><a href="mobiles.html">Mobiles</a></li>
      <li><a href="laptops.html">Laptops</a></li>
      <li><a href="tv.html">TV</a></li>
    </ul>
  </body>
  </html>
```

4.checkout.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Checkout - Durgesh Store</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <script>
    function highlightElectronics() {
      document.getElementById("electronics-link").style.color = "green";
    }
    window.onload = function() {
      document.getElementById("electronics-link").addEventListener("click", highlightElectronics);
    }
  </script>
</head>
<body>

  <nav>
    <a href="index.html">Home</a> |
    <a href="electronics.html" id="electronics-link">Electronics</a> |
    <a href="checkout.html">Checkout</a>
  </nav>

  <p>Home > Checkout</p>

  <h1>Checkout Form</h1>

  <form>

    <label for="name">Full Name:</label>
    <input type="text" id="name" name="name" placeholder="Enter your name" required>
    <br><br>

    <label for="address">Address:</label>
    <textarea id="address" name="address" placeholder="Enter your address" required></textarea>
    <br><br>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email" placeholder="Enter your email" required>
    <br><br>

    <label for="payment">Payment Method:</label>
    <input list="methods" id="payment" name="payment" placeholder="Choose payment method"
required>
    <datalist id="methods">
```

```
        <option value="UPI">
        <option value="Card">
        <option value="Cash">
        <option value="NetBanking">
    </datalist>
    <br><br>

    <label for="date">Delivery Date:</label>
    <input type="date" id="date" name="date" required>
    <br><br>

    <label for="coupon">Coupon Code:</label>
    <input type="text" id="coupon" name="coupon" placeholder="SHOP123" pattern="SHOP[0-
9]{3}" required>
    <br><br>

    <input type="submit" value="Place Order">
  </form>
</body>
</html>
```

CSS

Problem 1: Product Feature List Section (Level-1)

```html
<!DOCTYPE html>
<html>
<head>
  <title>Product Features</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <style>

    body {
      font-family: Arial, sans-serif;
      background-color: #f9f9f9;
      color: #333;
    }

    .features {
      list-style-type: none;
      padding: 0;
    }

    .features li {
      background-color: #fff;
      border: 2px solid #007acc;
      padding: 15px;
      margin: 10px 0;
      font-size: 18px;
      transition: background-color 0.3s, color 0.3s;
    }

    .features li:hover {
      background-color: #007acc;
      color: #fff;
    }

    h2 {
      font-family: "Trebuchet MS", sans-serif;
      color: #007acc;
    }
  </style>
</head>
<body>
  <section>
    <h2>Key Product Features</h2>
    <ul class="features">
      <li>High Performance and Speed</li>
      <li>User-Friendly Interface</li>
      <li>Secure and Reliable</li>
      <li>Cross-Platform Compatibility</li>
      <li>24/7 Customer Support</li>
    </ul>
  </section>
```

```
</body>
</html>
```

2. Responsive Blog Layout

```
<!DOCTYPE html>
<html>
<head>
  <title>Responsive Blog Homepage</title>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <style>

    body {
      font-family: Arial, sans-serif;
      margin: 0;
      background-color: #f9f9f9;
      color: #333;
    }

    header {
      background-color: #007acc;
      color: #fff;
      padding: 20px;
      text-align: center;
    }

    main {
      display: grid;
      grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
      gap: 20px;
      padding: 20px;
    }

    .card {
      background-color: #fff;
      border: 1px solid #ddd;
      padding: 20px;
      box-sizing: border-box;
      border-radius: 5px;
      transition: transform 0.3s;
    }

    .card:hover {
      transform: translateY(-5px);
      border-color: #007acc;
    }
```

```
        .card h2 {
            color: #007acc;
            margin-top: 0;
        }

        .card p {
            font-size: 15px;
            line-height: 1.5;
        }

        .card button {
            background-color: #007acc;
            color: #fff;
            border: none;
            padding: 10px 15px;
            cursor: pointer;
            border-radius: 3px;
            margin-top: 10px;
        }

        .card button:hover {
            background-color: #005f99;
        }

        @media (max-width: 768px) {
            main {
                grid-template-columns: 1fr;
            }
        }
    </style>
</head>
<body>
    <header>
        <h1>Durgesh's Tech Blog</h1>
        <p>Insights, tutorials, and stories from the world of technology</p>
    </header>

    <main>
        <div class="card">
            <h2>Understanding Flexbox</h2>
            <p>Learn how Flexbox simplifies layout design and makes responsive web pages easier to
build.</p>
            <button>Read More</button>
        </div>

        <div class="card">
            <h2>Getting Started with Python</h2>
```

```
        <p>A beginner-friendly guide to Python programming, covering basics and practical
examples.</p>
        <button>Read More</button>
      </div>

      <div class="card">
        <h2>Data Visualization Tips</h2>
        <p>Discover best practices for creating clear and impactful charts using Matplotlib and
Seaborn.</p>
        <button>Read More</button>
      </div>

      <div class="card">
        <h2>HTML5 Semantic Tags</h2>
        <p>Improve accessibility and SEO by using semantic tags like &lt;header&gt;, &lt;section&gt;,
and &lt;footer&gt;.</p>
        <button>Read More</button>
      </div>
    </main>
</body>
</html>
```

3. Responsive Dashboard Layout

```
<!DOCTYPE html>
<html>
<head>
    <title>Admin Dashboard</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <style>

      body {
        margin: 0;
        font-family: Arial, sans-serif;
      }

      .container {
        display: flex;
        min-height: 100vh;
      }

      .sidebar {
        background-color: #2c3e50;
        color: #fff;
        padding: 20px;
        width: 20%;
      }
```

```
.sidebar h2 {
   margin-top: 0;
}

.sidebar a {
   display: block;
   color: #fff;
   text-decoration: none;
   margin: 10px 0;
}

.sidebar a:hover {
   text-decoration: underline;
}

.main {
   flex: 1;
   padding: 20px;
   background-color: #ecf0f1;
   display: grid;
   grid-template-columns: repeat(auto-fit, minmax(250px, 1fr));
   gap: 20px;
}


.card {
   background-color: #fff;
   border: 1px solid #ccc;
   padding: 20px;
   box-sizing: border-box;
   border-radius: 5px;
}

.card h3 {
   margin-top: 0;
   color: #2c3e50;
}

@media (max-width: 768px) {
   .container {
      flex-direction: column;
   }

   .sidebar {
      width: 100%;
      display: flex;
      justify-content: space-around;
   }
```

```
        .sidebar a {
            margin: 0 10px;
        }

        .main {
            grid-template-columns: 1fr;
        }
    }
  </style>
</head>
<body>
  <div class="container">

    <aside class="sidebar">
        <h2>Dashboard</h2>
        <a href="#">Home</a>
        <a href="#">Users</a>
        <a href="#">Reports</a>
        <a href="#">Settings</a>
    </aside>

    <section class="main">
        <div class="card">
            <h3>Total Users</h3>
            <p>1,250</p>
        </div>
        <div class="card">
            <h3>Active Sessions</h3>
            <p>320</p>
        </div>
        <div class="card">
            <h3>Revenue</h3>
            <p>₹75,000</p>
        </div>
        <div class="card">
            <h3>Pending Tasks</h3>
            <p>18</p>
        </div>
    </section>
  </div>
</body>
</html>
```

Problem 4: Sticky Navbar + Active Link Highlight

```
<!DOCTYPE html>
<html>
<head>
  <title>Sticky Navbar Example</title>
  <style>
    nav {
      position: sticky; top: 0;
      background:#007acc; padding:15px;
      box-shadow:0 2px 6px rgba(0,0,0,0.3);
    }
    nav a {color:#fff; text-decoration:none; padding:8px 14px;}
    nav a:hover {background:#005f99;}
    .active {background:#004466;}
  </style>
</head>
<body>
  <nav>
    <a href="#" class="active">Home</a>
    <a href="#">About</a>
    <a href="#">Services</a>
    <a href="#">Blog</a>
    <a href="#">Contact</a>
  </nav>
  <p style="height:1500px;">Scroll down to see sticky navbar.</p>
</body>
</html>
```

Problem 5: Responsive Pricing Table (Flexbox)

```
<!DOCTYPE html>
<html>
<head>
  <style>
    nav {
      position: sticky; top: 0;
      background:#007acc; padding:10px;
      box-shadow:0 2px 5px rgba(0,0,0,0.3);
    }
    nav a {
      color:#fff; text-decoration:none; padding:8px 12px;
    }
    nav a:hover {background:#005f99;}
    .active {background:#004466;}
  </style>
```

```
</head>
<body>
  <nav>
    <a href="#" class="active">Home</a>
    <a href="#">About</a>
    <a href="#">Services</a>
    <a href="#">Blog</a>
    <a href="#">Contact</a>
  </nav>
  <p style="height:2000px;">Scroll to see sticky navbar...</p>
</body>
</html>
```

Problem 6: Grid Photo Gallery (Advanced Grid)

```
<!DOCTYPE html>
<html>
<head>
  <style>
    .pricing {display:flex;gap:20px;flex-wrap:wrap;}
    .card {
      flex:1;min-width:250px;background:#fff;
      border:1px solid #ccc;padding:20px;text-align:center;
      transition:transform 0.3s;
    }
    .card:hover {transform:scale(1.05);}
    @media(max-width:768px){.pricing{flex-direction:column;}}
  </style>
</head>
<body>
  <section class="pricing">
    <div class="card"><h2>Basic</h2><p>$10/mo</p><ul><li>Feature A</li><li>Feature
B</li></ul><button>Buy</button></div>
    <div class="card"><h2>Pro</h2><p>$20/mo</p><ul><li>Feature A</li><li>Feature
B</li></ul><button>Buy</button></div>
    <div class="card"><h2>Premium</h2><p>$30/mo</p><ul><li>Feature A</li><li>Feature
B</li></ul><button>Buy</button></div>
  </section>
</body>
</html>
```

Problem 7: Sidebar Layout with Hide/Show on Mobile

```html
<!DOCTYPE html>
<html>
<head>
  <title>Sidebar Layout</title>
  <style>
    .container {display:flex; min-height:100vh;}
    .sidebar {background:#2c3e50; color:#fff; padding:20px; width:200px;}
    .sidebar a {display:block; color:#fff; padding:8px; text-decoration:none;}
    .sidebar a:hover {background:#444;}
    .main {flex:1; padding:20px; background:#ecf0f1;}
    @media(max-width:768px){
      .container{flex-direction:column;}
      .sidebar{width:100%; display:flex; justify-content:space-around;}
      .main{width:100%;}
    }
  </style>
</head>
<body>
  <header style="background:#007acc;color:#fff;padding:10px;text-align:center;">Header</header>
  <div class="container">
    <aside class="sidebar">
      <a href="#">Home</a><a href="#">Users</a><a href="#">Reports</a><a href="#">Settings</a>
    </aside>
    <section class="main"><h2>Main Content</h2><p>Responsive sidebar demo.</p></section>
  </div>
  <footer style="background:#007acc;color:#fff;padding:10px;text-align:center;">Footer</footer>
</body>
</html>
```

Day 3(18-02-2026 to 19-02-2026)

Problem 1: ATM Withdrawal Program (if-else + operators)

```
function atmWithdrawal(balance, amount) {
   if (amount > balance) {
      console.log("Insufficient Balance");
   } else if (amount <= 0) {
      console.log("Invalid Amount");
   } else if (amount % 100 === 0) {
      balance -= amount;
      console.log("Withdrawal Successful! Remaining Balance: " + balance);
   } else {
      console.log("Enter amount in multiples of 100");
   }
}

let accountBalance = 5000;
let withdrawalAmount = 1200;

atmWithdrawal(accountBalance, withdrawalAmount);
```

Problem 2: Electricity Bill Calculator (if-else + loops optional)

```
function calculateBill(units) {
   let rate, bill;

   if (units <= 100) {
      rate = 2;
   } else if (units <= 200) {
      rate = 3;
   } else if (units <= 300) {
      rate = 5;
   } else {
      rate = 8;
   }

   bill = units * rate;

   console.log("Units Consumed: " + units);
   console.log("Total Bill Amount: ₹" + bill);
}

calculateBill(250);
```

Problem 3: Weekday Finder using Switch (switch case)

```
function findDay(dayNumber) {
   switch(dayNumber) {
      case 1:
         console.log("Monday");
         console.log("Weekday");
         break;
      case 2:
         console.log("Tuesday");
         console.log("Weekday");
         break;
      case 3:
         console.log("Wednesday");
         console.log("Weekday");
         break;
      case 4:
         console.log("Thursday");
         console.log("Weekday");
         break;
      case 5:
         console.log("Friday");
         console.log("Weekday");
         break;
      case 6:
         console.log("Saturday");
         console.log("Weekend");
         break;
      case 7:
         console.log("Sunday");
         console.log("Weekend");
         break;
      default:
         console.log("Invalid day number");
   }
}
findDay(6);
findDay(3);
findDay(9);
```

Problem 4: Password Validation Program (if-else + logical operators)

```
function validatePassword(password) {
   let hasDigit = false;
   let hasUpper = false;

   for (let i = 0; i < password.length; i++) {
      let ch = password[i];
      if (!isNaN(ch)) hasDigit = true;
      if (ch >= 'A' && ch <= 'Z') hasUpper = true;
   }
   if (password.length < 8) {
      console.log("Password must be at least 8 characters long");
   } else if (!hasDigit) {
      console.log("Password must contain a digit");
   } else if (!hasUpper) {
      console.log("Password must contain an uppercase letter");
   } else {
      console.log("Strong Password");
   }
}

validatePassword("Pass123");
validatePassword("password123");
validatePassword("Password");
validatePassword("Password123");
```

Problem 5: Traffic Signal Simulator (Level-2)

```
function trafficSignal(color) {
   switch(color) {
      case "red":
         console.log("Stop");
         break;
      case "yellow":
         console.log("Get Ready");
         break;
      case "green":
         console.log("Go");
         break;
      default:
         console.log("Invalid signal color");
   }
```

```
}

trafficSignal("red");
trafficSignal("yellow");
trafficSignal("green");
trafficSignal("blue");
```

Problem 6: Number Analysis Tool (Level-2)

```
function analyzeNumber(num) {
   let positivity = (num >= 0) ? "Positive" : "Negative";
   console.log("Number: " + num);
   console.log("Positivity: " + positivity);

   if (num % 2 === 0) {
      console.log("Parity: Even");
   } else {
      console.log("Parity: Odd");
   }
   console.log("Numbers from 1 to " + num + ":");
   for (let i = 1; i <= num; i++) {
      console.log(i);
   }
}
analyzeNumber(7);
```

Problem 7: Dynamic Student Profile Manager (Level-2)

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Profile System</title>
  <style>
    #profile {
      border: 2px solid #007acc;
      padding: 15px;
      margin: 20px;
      border-radius: 8px;
      background: #f0f8ff;
      width: 300px;
      font-family: Arial, sans-serif;
    }
    #profile h2 {
```

```
      color: #007acc;
      margin-top: 0;
    }
  </style>
</head>
<body>
  <div id="profile"></div>

  <button onclick="updateMarks(95)">Update Marks to 95</button>
  <button onclick="updateMarks(80)">Update Marks to 80</button>

  <script>
    let student = {
      name: "Durgesh",
      rollNo: 101,
      marks: 85
    };
    function updateStudentProfile(studentObj) {
      let profileDiv = document.getElementById("profile");
      profileDiv.innerHTML = `
        <h2>Student Profile</h2>
        <p><strong>Name:</strong> ${studentObj.name}</p>
        <p><strong>Roll No:</strong> ${studentObj.rollNo}</p>
        <p><strong>Marks:</strong> ${studentObj.marks}</p>
      `;
    }
    function updateMarks(newMarks) {
      student.marks = newMarks;
      updateStudentProfile(student);
    }
    updateStudentProfile(student);
  </script>
</body>
</html>
```

Problem 8: Discount Calculator (Function + Parameters + DOM)

```
<!DOCTYPE html>
<html>
<head>
  <title>Discount Calculator</title>
  <meta charset="UTF-8">
  <style>
```

```
    body {font-family: Arial, sans-serif; margin: 20px;}
    input {margin: 5px; padding: 8px;}
    button {padding: 8px 12px; background:#007acc; color:#fff; border:none; cursor:pointer;}
    button:hover {background:#005f99;}
    #result {margin-top: 15px; font-weight: bold; color:#007acc;}
  </style>
</head>
<body>
  <h2>Product Discount Calculator</h2>

  <label>Product Price: </label>
  <input type="number" id="price" placeholder="Enter price"><br>

  <label>Discount Percentage: </label>
  <input type="number" id="discount" placeholder="Enter discount %"><br>

  <button onclick="showDiscount()">Calculate</button>

  <p id="result"></p>

  <script>
    function calculateDiscount(price, discount) {
      let finalPrice = price - (price * discount / 100);
      return finalPrice;
    }
    function showDiscount() {
      let price = parseFloat(document.getElementById("price").value);
      let discount = parseFloat(document.getElementById("discount").value);

      if (isNaN(price) || isNaN(discount)) {
        document.getElementById("result").innerText = "Please enter valid numbers.";
        return;
      }

      let result = calculateDiscount(price, discount);
      document.getElementById("result").innerText = "Final Price after discount: ₹" +
result.toFixed(2);
    }
  </script>
</body>
</html>
```

Problem 9: Filter Passing Marks (filter() + DOM)

```
<!DOCTYPE html>
<html>
<head>
  <title>Filter Passing Marks</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; margin: 20px;}
    #output {border: 2px solid #007acc; padding: 15px; border-radius: 8px;
background:#f0f8ff;}
    h2 {color:#007acc;}
    p {margin:5px 0;}
  </style>
</head>
<body>
  <h2>Student Marks Analysis</h2>
  <div id="output"></div>
  <script>
    let marks = [35, 70, 88, 45, 92, 60, 25];
    let passedMarks = marks.filter(m => m >= 50);
    let passedCount = passedMarks.length;
    let outputDiv = document.getElementById("output");
    outputDiv.innerHTML = `
      <p><strong>Original Marks:</strong> ${marks.join(", ")}</p>
      <p><strong>Passed Marks (>=50):</strong> ${passedMarks.join(", ")}</p>
      <p><strong>Total Students Passed:</strong> ${passedCount}</p>
    `;
  </script>
</body>
</html>
```

Problem 10: Employee Object System (Objects + Functions)

```
<!DOCTYPE html>
<html>
<head>
  <title>Employee Object System</title>
  <meta charset="UTF-8">
  <style>
    #employeeProfile {
      border: 2px solid #007acc;
      padding: 15px;
      margin: 20px;
      border-radius: 8px;
      background: #f0f8ff;
      width: 350px;
```

```
    font-family: Arial, sans-serif;
  }
  #employeeProfile h2 {
    color: #007acc;
    margin-top: 0;
  }
  button {
    margin: 5px;
    padding: 8px 12px;
    background:#007acc;
    color:#fff;
    border:none;
    border-radius:4px;
    cursor:pointer;
  }
  button:hover {background:#005f99;}
 </style>
</head>
<body>
 <h2>Employee Profile System</h2>
 <div id="employeeProfile"></div>

 <button onclick="increaseSalary(employee, 10)">Increase Salary by 10%</button>
 <button onclick="increaseSalary(employee, 30)">Increase Salary by 30%</button>

 <script>
  let employee = {
    name: "Amit",
    salary: 40000,
    department: "IT"
  };
  function increaseSalary(emp, percentage) {
    emp.salary = emp.salary + (emp.salary * percentage / 100);
    updateEmployeeProfile(emp);
    return emp.salary;
  }
  function updateEmployeeProfile(emp) {
    let status = (emp.salary > 50000) ? "Senior Employee" : "Junior Employee";
    document.getElementById("employeeProfile").innerHTML = `
      <h2>Employee Details</h2>
      <p><strong>Name:</strong> ${emp.name}</p>
      <p><strong>Department:</strong> ${emp.department}</p>
      <p><strong>Salary:</strong> ₹${emp.salary}</p>
      <p><strong>Status:</strong> ${status}</p> }
```

```
    updateEmployeeProfile(employee);
  </script>
</body>
</html>
```

Problem 11: Scope Debug Program (Global vs Local vs Block)

```
<!DOCTYPE html>
<html>
<head>
  <title>Scope Demo</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; margin: 20px;}
    #output {border: 2px solid #007acc; padding: 15px; border-radius: 8px;
background:#f0f8ff;}
  </style>
</head>
<body>
  <h2>Scope Understanding Demo</h2>
  <div id="output"></div>

  <script>
    var globalVar = "I am Global";

    function testScope() {
      var functionVar = "I am Function Scoped";
      {
        let blockVar = "I am Block Scoped";
        console.log(globalVar);
        console.log(functionVar);
        console.log(blockVar);

        document.getElementById("output").innerHTML += `
          <p>Global Variable: ${globalVar}</p>
          <p>Function Variable: ${functionVar}</p>
          <p>Block Variable (inside block): ${blockVar}</p>
      }
      try {
        console.log(blockVar);
        document.getElementById("output").innerHTML += `<p>Block Variable (outside block):
${blockVar}</p>`;
      } catch (error) {
        console.log("Error accessing blockVar outside block:", error.message);
```

```
      document.getElementById("output").innerHTML += `<p style="color:red;">Error:
Cannot access blockVar outside block scope</p>`;
     }
   }
   testScope();
  </script>
</body>
</html>
```

Problem 12: DOM Color Theme Changer (Events + DOM)

```
<!DOCTYPE html>
<html>
<head>
  <title>Theme Switcher</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; text-align: center; padding: 30px;}
    h2 {margin-bottom: 20px;}
    button {
      margin: 5px; padding: 10px 15px;
      border: none; border-radius: 5px;
      cursor: pointer; font-size: 14px;
    }
    #mode {margin-top: 20px; font-weight: bold;}
  </style>
</head>
<body>
  <h2 id="heading">Theme Switcher Demo</h2>

  <button onclick="setDarkMode()">Dark Mode</button>
  <button onclick="setLightMode()">Light Mode</button>
  <button onclick="setBlueMode()">Blue Mode</button>

  <p id="mode">Current Mode: Light Mode</p>

  <script>
    function setDarkMode() {
      document.body.style.backgroundColor = "#222";
      document.body.style.color = "#fff";
      document.getElementById("heading").style.color = "#ffcc00";
      document.getElementById("mode").innerText = "Current Mode: Dark Mode";
    }

    function setLightMode() {
```

```
    document.body.style.backgroundColor = "#fff";
    document.body.style.color = "#000";
    document.getElementById("heading").style.color = "#007acc";
    document.getElementById("mode").innerText = "Current Mode: Light Mode";
  }

  function setBlueMode() {
    document.body.style.backgroundColor = "#007acc";
    document.body.style.color = "#fff";
    document.getElementById("heading").style.color = "#ffeb3b";
    document.getElementById("mode").innerText = "Current Mode: Blue Mode";
  }
 </script>
</body>
</html>
```

Problem 13: Search Numbers Greater than Input (Function + filter + User Input)

```
<!DOCTYPE html>
<html>
<head>
 <title>Filter Numbers</title>
 <meta charset="UTF-8">
 <style>
  body {font-family: Arial, sans-serif; margin: 20px;}
  input, button {margin: 5px; padding: 8px;}
  #result {margin-top: 15px; font-weight: bold; color:#007acc;}
 </style>
</head>
<body>
 <h2>Filter Numbers Greater Than Input</h2>

 <label>Enter a value: </label>
 <input type="number" id="userValue" placeholder="Enter number">
 <button onclick="filterNumbers()">Filter</button>

 <div id="result"></div>

 <script>
  let numbers = [10, 25, 60, 75, 90, 15, 45, 120];
  function filterNumbers() {
    let inputVal = parseFloat(document.getElementById("userValue").value);
    if (isNaN(inputVal)) {
```

```
      document.getElementById("result").innerText = "Please enter a valid number.";
      return;
    }
    let filtered = numbers.filter(num => num > inputVal);
    filtered.sort((a, b) => a - b);
    document.getElementById("result").innerHTML = `
      <p><strong>Original Numbers:</strong> ${numbers.join(", ")}</p>
      <p><strong>Filtered Numbers (greater than ${inputVal}):</strong> ${filtered.join(",
")}</p>
      <p><strong>Total Count:</strong> ${filtered.length}</p>
  }
 </script>
</body>
</html>
```

Day 5(20-02-2026)

Problem 1: Location-Based Weather Logger (Level-2)

```
<!DOCTYPE html>
<html>
<head>
  <title>Geolocation Tracker</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; margin: 20px;}
    button {
      padding: 10px 15px; background:#007acc; color:#fff;
      border:none; border-radius:5px; cursor:pointer;
    }
    button:hover {background:#005f99;}
    #output {margin-top:20px;}
    #history {margin-top:20px; border:1px solid #ccc; padding:10px; border-radius:5px;}
    h3 {color:#007acc;}
  </style>
</head>
<body>
  <h2>Geolocation Tracker</h2>
  <button onclick="getMyLocation()">Get My Location</button>

  <div id="output"></div>
  <div id="history">
    <h3>Location History (Last 5)</h3>
    <ul id="historyList"></ul>
  </div>

  <script>
    function getMyLocation() {
      if (navigator.geolocation) {
        navigator.geolocation.getCurrentPosition(successCallback, errorCallback, {
          timeout: 10000
        });
      } else {
        document.getElementById("output").innerText = "Geolocation not supported.";
      }
    }
    function successCallback(position) {
      let lat = position.coords.latitude;
      let lon = position.coords.longitude;
      document.getElementById("output").innerHTML = `
```

```
      <p><strong>Latitude:</strong> ${lat}</p>
      <p><strong>Longitude:</strong> ${lon}</p>
    `;

    saveLocation(lat, lon);
    displayHistory();
  }
  function errorCallback(error) {
    let msg;
    switch(error.code) {
      case error.PERMISSION_DENIED:
        msg = "Permission denied.";
        break;
      case error.TIMEOUT:
        msg = "Request timed out.";
        break;
      case error.POSITION_UNAVAILABLE:
        msg = "Location unavailable.";
        break;
      default:
        msg = "Unknown error.";
    }
    document.getElementById("output").innerText = msg;
  }
  function saveLocation(lat, lon) {
    let history = JSON.parse(localStorage.getItem("locationHistory")) || [];
    history.unshift({latitude: lat, longitude: lon, time: new Date().toLocaleString()});
    if (history.length > 5) history.pop();
    localStorage.setItem("locationHistory", JSON.stringify(history));
  }
  function displayHistory() {
    let history = JSON.parse(localStorage.getItem("locationHistory")) || [];
    let list = document.getElementById("historyList");
    list.innerHTML = "";
    history.forEach(entry => {
      let li = document.createElement("li");
      li.textContent = `Lat: ${entry.latitude}, Lon: ${entry.longitude} (Time: ${entry.time})`;
      list.appendChild(li);
    });
  }
  window.onload = displayHistory;
 </script>
</body>
</html>
```

Problem 2: Mini Expense Tracker using Client-Side Database (Level-2)

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Mini Expense Tracker</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; margin: 20px;}
    input, button {margin: 5px; padding: 8px;}
    table {border-collapse: collapse; margin-top: 15px; width: 100%;}
    th, td {border: 1px solid #ccc; padding: 8px; text-align: center;}
    th {background: #007acc; color: #fff;}
    button {background:#007acc; color:#fff; border:none; cursor:pointer;}
    button:hover {background:#005f99;}
  </style>
</head>
<body>
  <h2>Mini Expense Tracker (Client-Side DB)</h2>
  <label>Expense Title:</label>
  <input type="text" id="title"><br>
  <label>Amount:</label>
  <input type="number" id="amount"><br>
  <label>Date:</label>
  <input type="date" id="date"><br>
  <button onclick="addExpense()">Add Expense</button>
  <button onclick="viewExpenses()">View Expenses</button>
  <div id="output"></div>
  <script src="app.js"></script>
</body>
</html>
```

App.js

```
let db = openDatabase("ExpenseDB", "1.0", "Expense Tracker DB", 2 * 1024 * 1024);
db.transaction(function(tx) {
  tx.executeSql("CREATE TABLE IF NOT EXISTS expenses (id INTEGER PRIMARY KEY, title TEXT,
amount REAL, date TEXT)");
});
function addExpense() {
  let title = document.getElementById("title").value;
  let amount = document.getElementById("amount").value;
  let date = document.getElementById("date").value;
  if (!title || !amount || !date) {
    alert("Please fill all fields!");
```

```
    return;
  }
  db.transaction(function(tx) {
    tx.executeSql("INSERT INTO expenses (title, amount, date) VALUES (?, ?, ?)",
      [title, amount, date],
      function() { alert("Expense Added!"); },
      function(tx, error) { alert("Error: " + error.message); }
    );
  });
}
function viewExpenses() {
  db.transaction(function(tx) {
    tx.executeSql("SELECT * FROM expenses", [], function(tx, results) {
      let len = results.rows.length;
      let output =
"<table><tr><th>ID</th><th>Title</th><th>Amount</th><th>Date</th><th>Action</th></t
r>";
      for (let i = 0; i < len; i++) {
        let row = results.rows.item(i);
        output += `<tr>
          <td>${row.id}</td>
          <td>${row.title}</td>
          <td>₹${row.amount}</td>
          <td>${row.date}</td>
          <td><button onclick="deleteExpense(${row.id})">Delete</button></td>
        </tr>`;
      }
      output += "</table>";
      document.getElementById("output").innerHTML = output;
    });
  });
}
function deleteExpense(id) {
  db.transaction(function(tx) {
    tx.executeSql("DELETE FROM expenses WHERE id=?", [id],
      function() { alert("Expense Deleted!"); viewExpenses(); },
      function(tx, error) { alert("Error: " + error.message); }
    );
  });
}
window.onload = viewExpenses;
```

Problem 3: Attendance Tracker System (IndexedDB + Events + SessionStorage)

Index.html

```html
<!DOCTYPE html>
<html>
<head>
  <title>Attendance Tracker</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; margin: 20px;}
    input, select, button {margin: 5px; padding: 8px;}
    table {border-collapse: collapse; margin-top: 15px; width: 100%;}
    th, td {border: 1px solid #ccc; padding: 8px; text-align: center;}
    th {background: #007acc; color: #fff;}
    button {background:#007acc; color:#fff; border:none; cursor:pointer;}
    button:hover {background:#005f99;}
    #lastName {margin-top: 10px; font-weight: bold; color:#007acc;}
  </style>
</head>
<body>
  <h2>Attendance Tracker</h2>

  <label>Student Name:</label>
  <input type="text" id="studentName" required><br>

  <label>Date:</label>
  <input type="date" id="date" required><br>

  <label>Status:</label>
  <select id="status">
    <option value="Present">Present</option>
    <option value="Absent">Absent</option>
  </select><br>

  <button onclick="saveAttendance()">Save Attendance</button>
  <button onclick="showAttendance()">Show Attendance</button>
  <button onclick="clearAttendance()">Clear Attendance Database</button>

  <div id="lastName"></div>
  <div id="output"></div>

  <script src="attendance.js"></script>
</body>
</html>
```

Attendance.js

```
let db;
let request = indexedDB.open("AttendanceDB", 1);
request.onupgradeneeded = function(event) {
  db = event.target.result;
  let store = db.createObjectStore("attendance", { keyPath: "id", autoIncrement: true });
  store.createIndex("name", "name", { unique: false });
  store.createIndex("date", "date", { unique: false });
  store.createIndex("status", "status", { unique: false });
};
request.onsuccess = function(event) {
  db = event.target.result;
  showAttendance();
};

request.onerror = function(event) {
  console.error("Database error:", event.target.errorCode);
};
function saveAttendance() {
  let name = document.getElementById("studentName").value;
  let date = document.getElementById("date").value;
  let status = document.getElementById("status").value;
  if (!name || !date) {
    alert("Please fill all required fields!");
    return;
  }
  let transaction = db.transaction(["attendance"], "readwrite");
  let store = transaction.objectStore("attendance");
  let record = { name, date, status };
  store.add(record);
  transaction.oncomplete = function() {
    alert("Attendance Saved!");
    sessionStorage.setItem("lastName", name);
    document.getElementById("lastName").innerText = "Last Submitted Student: " + name;
    showAttendance();
  };

  transaction.onerror = function(event) {
    alert("Error saving attendance: " + event.target.error);
  };
}
function showAttendance() {
  let transaction = db.transaction(["attendance"], "readonly");
  let store = transaction.objectStore("attendance");
```

```
  let request = store.getAll();

  request.onsuccess = function(event) {
    let records = event.target.result;
    let output =
"<table><tr><th>ID</th><th>Name</th><th>Date</th><th>Status</th></tr>";
    records.forEach(r => {
      output += `<tr>
        <td>${r.id}</td>
        <td>${r.name}</td>
        <td>${r.date}</td>
        <td>${r.status}</td>
      </tr>`;
    });
    output += "</table>";
    document.getElementById("output").innerHTML = output;
    let lastName = sessionStorage.getItem("lastName");
    if (lastName) {
      document.getElementById("lastName").innerText = "Last Submitted Student: " +
lastName;
    }
  };
}
function clearAttendance() {
  let transaction = db.transaction(["attendance"], "readwrite");
  let store = transaction.objectStore("attendance");
  let clearReq = store.clear();
  clearReq.onsuccess = function() {
    alert("Attendance Database Cleared!");
    document.getElementById("output").innerHTML = "";
    sessionStorage.removeItem("lastName");
    document.getElementById("lastName").innerText = "";
  };
}
```

Problem 4: Delivery Location Saver (Geolocation + localStorage + IndexedDB)

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Delivery App - Location Tracker</title>
  <meta charset="UTF-8">
  <style>
```

```
    body {font-family: Arial, sans-serif; margin: 20px;}
    button {
      margin: 5px; padding: 10px 15px;
      background:#007acc; color:#fff; border:none; border-radius:5px;
      cursor:pointer;
    }
    button:hover {background:#005f99;}
    #output, #history {margin-top: 20px;}
    table {border-collapse: collapse; width: 100%; margin-top: 10px;}
    th, td {border: 1px solid #ccc; padding: 8px; text-align: center;}
    th {background: #007acc; color: #fff;}
  </style>
</head>
<body>
  <h2>Delivery App - Location Tracker</h2>

  <button onclick="detectLocation()">Detect My Location</button>
  <button onclick="viewHistory()">View Location History</button>
  <div id="output"></div>
  <div id="history"></div>
  <script src="delivery.js"></script>
</body>
</html>
```

Delivery.js

```
let db;
let request = indexedDB.open("DeliveryDB", 1);

request.onupgradeneeded = function(event) {
  db = event.target.result;
  let store = db.createObjectStore("deliveryLocations", { keyPath: "id", autoIncrement: true
});
  store.createIndex("latitude", "latitude", { unique: false });
  store.createIndex("longitude", "longitude", { unique: false });
  store.createIndex("dateTime", "dateTime", { unique: false });
};
request.onsuccess = function(event) {
  db = event.target.result;
  viewHistory();
};
request.onerror = function(event) {
  console.error("Database error:", event.target.errorCode);
};
function detectLocation() {
```

```
  if (navigator.geolocation) {
    navigator.geolocation.getCurrentPosition(successCallback, errorCallback, { timeout: 10000
});
  } else {
    document.getElementById("output").innerText = "Geolocation not supported.";
  }
}
function successCallback(position) {
  let lat = position.coords.latitude;
  let lon = position.coords.longitude;
  let dateTime = new Date().toLocaleString();
  document.getElementById("output").innerHTML = `
    <p><strong>Latitude:</strong> ${lat}</p>
    <p><strong>Longitude:</strong> ${lon}</p>
    <p><strong>Time:</strong> ${dateTime}</p>
    localStorage.setItem("lastLocation", JSON.stringify({ latitude: lat, longitude: lon, dateTime
}));

  let transaction = db.transaction(["deliveryLocations"], "readwrite");
  let store = transaction.objectStore("deliveryLocations");
  store.add({ latitude: lat, longitude: lon, dateTime });
  transaction.oncomplete = function() {
    console.log("Location saved to IndexedDB");
    viewHistory();
  };
}
function errorCallback(error) {
  let msg;
  switch(error.code) {
    case error.PERMISSION_DENIED:
      msg = "Location access denied.";
      break;
    case error.TIMEOUT:
      msg = "Request timed out.";
      break;
    case error.POSITION_UNAVAILABLE:
      msg = "Location unavailable.";
      break;
    default:
      msg = "Unknown error.";
  }
  document.getElementById("output").innerText = msg;
}
function viewHistory() {
```

```
  let transaction = db.transaction(["deliveryLocations"], "readonly");
  let store = transaction.objectStore("deliveryLocations");
  let request = store.getAll();

  request.onsuccess = function(event) {
    let records = event.target.result;
    if (records.length === 0) {
      document.getElementById("history").innerHTML = "<p>No location history found.</p>";
      return;
    }

    let output = "<h3>Location
History</h3><table><tr><th>ID</th><th>Latitude</th><th>Longitude</th><th>DateTime</t
h></tr>";
    records.forEach(r => {
      output += `<tr>
        <td>${r.id}</td>
        <td>${r.latitude}</td>
        <td>${r.longitude}</td>
        <td>${r.dateTime}</td>
      </tr>`;
    });
    output += "</table>";
    document.getElementById("history").innerHTML = output;
  };
}
```

PROBLEM 1: Student Marks Analyzer

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Marks Analyzer</title>
  <meta charset="UTF-8">
</head>
<body>
  <h2>Student Marks Analyzer</h2>
  <div id="output"></div>
  <script type="module" src="marks.js"></script>
</body>
</html>
```

Marks.js

```
const marks = [78, 85, 92, 67, 54, 89];
const calculateTotal = arr => arr.reduce((sum, val) => sum + val, 0);
const calculateAverage = arr => calculateTotal(arr) / arr.length;
const checkResult = avg => avg >= 50 ? "Pass" : "Fail";
const displayResults = () => {
  const total = calculateTotal(marks);
  const average = calculateAverage(marks);
  const result = checkResult(average);
  document.getElementById("output").innerHTML = `
    <p><strong>Marks:</strong> ${marks.join(", ")}</p>
    <p><strong>Total Marks:</strong> ${total}</p>
    <p><strong>Average Marks:</strong> ${average.toFixed(2)}</p>
    <p><strong>Result:</strong> ${result}</p>
  `;
};
displayResults();
```

Problem 2: Student Result Analyzer (Arrays + Methods + Template Literals)

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Student Marks Report</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; margin: 20px;}
    button {
      margin: 5px; padding: 8px 12px;
      background:#007acc; color:#fff; border:none; border-radius:4px;
      cursor:pointer;
    }
    button:hover {background:#005f99;}
    table {border-collapse: collapse; margin-top: 15px; width: 100%;}
    th, td {border: 1px solid #ccc; padding: 8px; text-align: center;}
    th {background: #007acc; color: #fff;}
  </style>
</head>
<body>
  <h2>Student Marks Report</h2>
```

```
  <button onclick="sortStudents()">Sort Students by Marks</button>
  <button onclick="filterPassed()">Filter Passed Students</button>
  <button onclick="calculateAverage()">Calculate Average Marks</button>
  <button onclick="displayStudents()">Display All Students</button>

  <div id="output"></div>

  <script src="students.js"></script>
</body>
</html>
```

Student.js

```
let students = [
  { name: "Kunal", marks: 85 },
  { name: "Rahul", marks: 40 },
  { name: "Neha", marks: 72 },
  { name: "Amit", marks: 55 },
  { name: "Priya", marks: 30 }
];
const getGrade = marks =>
  marks >= 80 ? "A" :
  marks >= 60 ? "B" :
  marks >= 50 ? "C" : "Fail";
function displayStudents(list = students) {
  let output = "<table><tr><th>Name</th><th>Marks</th><th>Grade</th></tr>";
  list.map(s => {
    output += `<tr>
      <td>${s.name}</td>
      <td>${s.marks}</td>
      <td>${getGrade(s.marks)}</td>
    </tr>`;
  });
  output += "</table>";
  document.getElementById("output").innerHTML = output;
}
function sortStudents() {
  let sorted = [...students].sort((a, b) => b.marks - a.marks);
  displayStudents(sorted);
}
function filterPassed() {
  let passed = students.filter(s => s.marks >= 50);
  displayStudents(passed);
}
```

```
function calculateAverage() {
  const total = students.reduce((sum, s) => sum + s.marks, 0);
  const avg = total / students.length;
  document.getElementById("output").innerHTML = `
    <p><strong>Total Marks:</strong> ${total}</p>
    <p><strong>Average Marks:</strong> ${avg.toFixed(2)}</p>
    <p><strong>Result:</strong> ${avg >= 50 ? "Pass" : "Fail"}</p>
}
displayStudents();
```

Problem 3: Async Order Status Tracker (Callbacks + Promises + Async/Await)

Index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Order Delivery Tracking</title>
  <meta charset="UTF-8">
  <style>
    body {font-family: Arial, sans-serif; margin: 20px;}
    button {
      margin: 5px; padding: 8px 12px;
      background:#007acc; color:#fff; border:none; border-radius:4px;
      cursor:pointer;
    }
    button:hover {background:#005f99;}
    #output {margin-top: 20px; font-weight: bold;}
  </style>
</head>
<body>
  <h2>Order Delivery Tracking System</h2>
  <button onclick="trackOrderUsingCallback()">Track Order using Callback</button>
  <button onclick="trackOrderUsingPromise()">Track Order using Promise</button>
  <button onclick="trackOrderUsingAsync()">Track Order using Async/Await</button>
  <div id="output"></div>
  <script src="orders.js"></script>
</body>
</html>
```

Orders.js

```
const orders = [
  { orderId: 101, status: "Placed" },
  { orderId: 102, status: "Shipped" },
```

```
  { orderId: 103, status: "Delivered" }
];
function trackOrderCallback(orderId, callback) {
  setTimeout(() => {
    const order = orders.find(o => o.orderId === orderId);
    if (order) {
      callback(null, order.status);
    } else {
      callback("Order not found", null);
    }
  }, 2000);
}
function trackOrderUsingCallback() {
  const orderId = parseInt(prompt("Enter Order ID to track (Callback):"));
  trackOrderCallback(orderId, (err, status) => {
    if (err) {
      document.getElementById("output").innerText = err;
    } else {
      document.getElementById("output").innerText = `Order ID: ${orderId}\nStatus:
${status}`;
    }
  });
}
function trackOrderPromise(orderId) {
  return new Promise((resolve, reject) => {
    setTimeout(() => {
      const order = orders.find(o => o.orderId === orderId);
      if (order) {
        resolve(order.status);
      } else {
        reject("Order not found");
      }
    }, 2000);
  });
}

function trackOrderUsingPromise() {
  const orderId = parseInt(prompt("Enter Order ID to track (Promise):"));
  trackOrderPromise(orderId)
    .then(status => {
      document.getElementById("output").innerText = `Order ID: ${orderId}\nStatus:
${status}`;
    })
    .catch(err => {
```

```
    document.getElementById("output").innerText = err;
  });
}
async function trackOrderAsync(orderId) {
  try {
    const status = await trackOrderPromise(orderId);
    document.getElementById("output").innerText = `Order ID: ${orderId}\nStatus: ${status}`;
  } catch (err) {
    document.getElementById("output").innerText = err;
  }
}

function trackOrderUsingAsync() {
  const orderId = parseInt(prompt("Enter Order ID to track (Async/Await):"));
  trackOrderAsync(orderId);
}
```